

Experimental methodology

MIT-BIH Arrhythmia DB
(48 recordings, 360 Hz)

R-peak 1-s windows, classes N/A/V
per-window normalization to $[-1, 1]$

Patient-level split
train / validation / test (record-aware)

Architectures: CNN . LSTM . GRU . TCN
+ Conv1D autoencoder denoiser

E4 Quantization + calibration
FP32 to int8 TFLite (CNN/TCN full-int8,
LSTM/GRU float32); temperature T

Experiments
E1 grouped CV . E2 noise 0-40 dB
E3 external SVDB validation

E5 On-device verification
ESP32-S3 int8/fp32, memory and latency